Imperial College London

Planning Report for MSc Project

Title: Investigating the Temporal Dynamics of EEG Signals in Sequential Sensorimotor Action

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1. Project Specification

1.1. Background

Multiple sensory stimuli are typically involved when executing an action. Research shows that the choice of action is dependent on these stimuli from the environment.[1] These sensorimotor actions involve coordinated neural processes across various regions of the brain, notably within primary motor, premotor, parietal, and temporal cortices.[2], [3] However, it remain unclear exactly how these regions are wired and work together, particularly in terms of the precise timing, sequencing, and connectivity of different brain regions during sensorimotor actions.

1.2. Objective, Research Questions, and Hypotheses

Therefore, this research will be aiming to further investigate the relationship between these cortical regions with the electroencephalogram (EEG), mainly in the temporal domain. The main approach will involve using neural "markers," such as event-related (de)synchronization (ERS/ERD), to identify if the region is activated at each timestamp. And then, find out the temporal sequence and temporal difference between cortical regions. With this temporal information, there are two main directions of research questions:

a) How do factors such as sequential actions, multisensory input and attention affect the temporal dynamics?

   Hypotheses:
   - H1: Multisensory input will result in earlier ERD onset and higher SSVEP/ASSR amplitudes compared to unisensory input, indicating enhanced and faster cortical activation.
   - H2: Sequential motor actions will show longer and more distributed ERD periods across motor and prefrontal regions compared to simple actions, reflecting increased cognitive planning.
   - H3: Increased attention will lead to more consistent and faster activation, improving the brain's response timing.

b) Will the temporal information be useful to improve the motor imagery (MI)-based Brain Computer Interface (BCI)?

   Hypotheses:
   - H4: Using the temporal features (i.e. ERD onset latency, duration from multiple brain regions and regional activation differences in time) will improve the performance of motor imagery (MI)-based BCI systems.

2. Ethical Analysis

The study involves adult human participants who volunteer to undergo EEG recording during tasks involving visual, auditory, and motor stimuli. Participants will be recruited from within the college community and the surrounding population. Informed consent is obtained after a thorough explanation of the procedures and potential risks, ensuring that each participant's autonomy is respected. No invasive procedures or physical specimens (e.g., tissue biopsies or blood samples) are collected, which further minimizes ethical concerns related to bodily integrity. Data collected during this study include EEG signals. All data will be anonymized after acquisition.

3. Literature Review

3.1. Multi-sensorimotor Action and Research Gap

Complex multi-sensorimotor actions like catching a ball, and riding a bike are commonly done in everyday life. Studies have consistently shown that sensory information could enhance the performance of these motor action [1]. Despite these observations in performance benefits, there is still a significant gap in understanding the underlying neural dynamic: the precise millisecond-level temporal sequencing and latency differences in cortical activations during the sensorimotor actions have not been sufficiently quantified.

Early studies, such as those by Driver and Noesselt [4], and Calvert et al. [5], primarily focused on the behavioural outcomes and spatial mapping of multisensory integration. However, more recent works have started to look into the temporal and spatiotemporal side of multisensory integration. For instance,, Senkowski and Engel emphasised how neural dynamics across multiple timescales allow the flexibility to integrate, segregate and select sensory information in different time windows [6]. Yu and Zhao highlight the potential of spatiotemporal neural dynamics to inform brain-inspired network models for cross-modal recognition [7]. Moreover, Thura et al. have provided crucial insights into sensorimotor decision-making and action execution: revealing that decision-making and action execution are interconnected processes governed by a unified neural system, with distinct contributions from different brain regions [8]. These findings highlight the need to capture detailed temporal dynamics, a need that traditional behavioural and spatial methods have not met.

3.2. Applications and Clinical Implications

Accurately mapping the timing of neural activity offers great potential for clinical and technological improvements. Quantifying temporal sequencing and latency differences in cortical activation can provide objective metrics. This metrics could potentially be applied to distinguish normal from pathological neural processing, thereby enhancing diagnostic accuracy for various neurological conditions. In MI-based BCIs, using these timing metrics might include information that can lead to better device control and overall performance.

3.3. Methodological Strategies

3.3.1. Quantifying Temporal Differences

The project focuses on measuring the timing difference between activation events in different parts of the cortex. By using reliable markers (i.e., ERS/ERD) with EEG data, the exact moments when each region becomes active could be identified. Building a clear timeline of these activations will show the order in which multisensory inputs are processed.

There are several possible choices for the markers in this pipeline. They can be grouped into two categories: a) General Cortical Markers, b) Evoked Sensory Responses

a) General EEG Activity Evaluation Method

These markers are not specified to a particular sensory region. Examples are:

ERS/ERD: Event-related synchronization (ERS) and event-related desynchronization (ERD) reflect power changes in specific EEG frequency bands in response to an event. Typically, ERD corresponds to a reduction in power, indicating cortical activation, while ERS refers to an increase in

power, possibly reflecting recovery or inhibitory processes after activation. To extract these patterns, researchers often employ time–frequency decomposition techniques such as the Short-Time Fourier Transform (STFT) or wavelet transform. These methods yield both temporal and spectral information, enabling precise localization of when and at what frequencies these changes occur [9].

Microstates: EEG microstates are brief, quasi-stable periods—lasting roughly 80–120 milliseconds—during which the spatial configuration of scalp potentials remains relatively constant. They are considered the "atoms of thought," reflecting the basic building blocks of cognitive processing. Typically, microstate analysis involves calculating the Global Field Power (GFP) to determine moments of maximum signal strength, followed by clustering algorithms (such as k-means) to identify distinct microstate classes. Standardized toolboxes and smoothing techniques are then used to segment the continuous EEG into these microstates [10].

b) Evoked Sensory Responses

These markers are specifically designed to assess activity in targeted sensory regions. Here are just a few examples of them:

Steady-State Visual Evoked Potentials (SSVEP): SSVEPs are periodic responses elicited by visual stimuli that flicker at a constant frequency. When a subject views such a stimulus, the visual cortex produces responses that are phase-locked to the stimulation frequency, along with its harmonics. Processing typically involves transforming the time-domain data into the frequency domain using methods like Fast Fourier Transform (FFT) or STFT to extract the response peaks. Spatial filtering techniques, such as Independent Component Analysis (ICA), may also be applied to improve the signal-to-noise ratio [11].

Auditory Steady-State Responses (ASSR): ASSRs are elicited by amplitude- or frequency-modulated auditory stimuli and reflect the brain's ability to follow rapid auditory changes. These responses are phase-locked to the modulation frequency and can be identified by applying Fourier analysis, which reveals a continuous signal at the stimulation frequency. Common processing techniques include averaging and narrow-band filtering to isolate the ASSR signal from background noise [12], [13].

3.3.2. EEG Electrode Placement for Data Acquisition

Accurate electrode placement is essential for capturing the neural signals required in a multimodal experiment, where stimuli span visual, auditory, and motor domains. Placement could be varied to focus on different regions (i.e. clustered placement near occipital region to locate the SSVEP signals). Here is just an example placement that designed to best capture both general cortical activity and region-specific responses, as summarized below:

Region	Electrodes	Markers Supported	Rationale
Frontal	Fp1, Fp2, F3, Fz, F4	Microstates, global ERS/ERD	Captures anterior brain activity; supports global dynamics and cognitive processing frameworks.
Central	C3, Cz, C4	ERS/ERD, MI-BCI	Essential for sensorimotor rhythms, indexing motor planning and execution during motor imagery tasks.

Temporal	T7, T8	ASSR	Targets auditory processing pathways, ensuring robust capture of responses to auditory stimuli.
Parietal	P3, Pz, P4	Microstates, supportive of global ERS/ERD	Enhances coverage for integrating sensory inputs and higher-order processing involved in multisensory integration.
Occipital	O1, Oz, O2	SSVEP	Focuses on the primary visual cortex, crucial for capturing responses to visual flicker stimuli.

Table 1: EEG electrode placement. Summary of the choice of electrodes, its corresponding region, rationale of choice, and the markers supported.

Central and Frontal Regions (C3, Cz, C4, Fp1, Fp2, F3, Fz, F4): These electrodes are vital for capturing motor-related signals (ERS/ERD, MI-BCI) and global cortical dynamics (microstates). Such configurations are extensively used in studies on motor imagery and sensorimotor integration [9], [14].

Temporal Sites (T7, T8): These sites focus on the auditory cortices and brainstem regions, allowing for effective capture of ASSR components [12], [13].

Parietal Regions (P3, Pz, P4): The parietal cortex is well documented for its role in multisensory integration and spatial attention. For instance, Barutchu et al. observed beta-range oscillatory shifts in parietal areas in response to multisensory stimuli [15], and work by Park and Kayser highlights the mediation of multisensory integration by parietal regions [16]. As such, including P3, Pz, and P4 enhances our ability to capture the neural dynamics associated with the integration of visual and auditory information, thereby complementing the central and occipital recordings.

Occipital Sites (O1, Oz, O2): These electrodes are standard for recording SSVEP responses, ensuring that the visual cortex is adequately sampled for processing flicker-evoked neural activity [11].

4. Implementation Plan

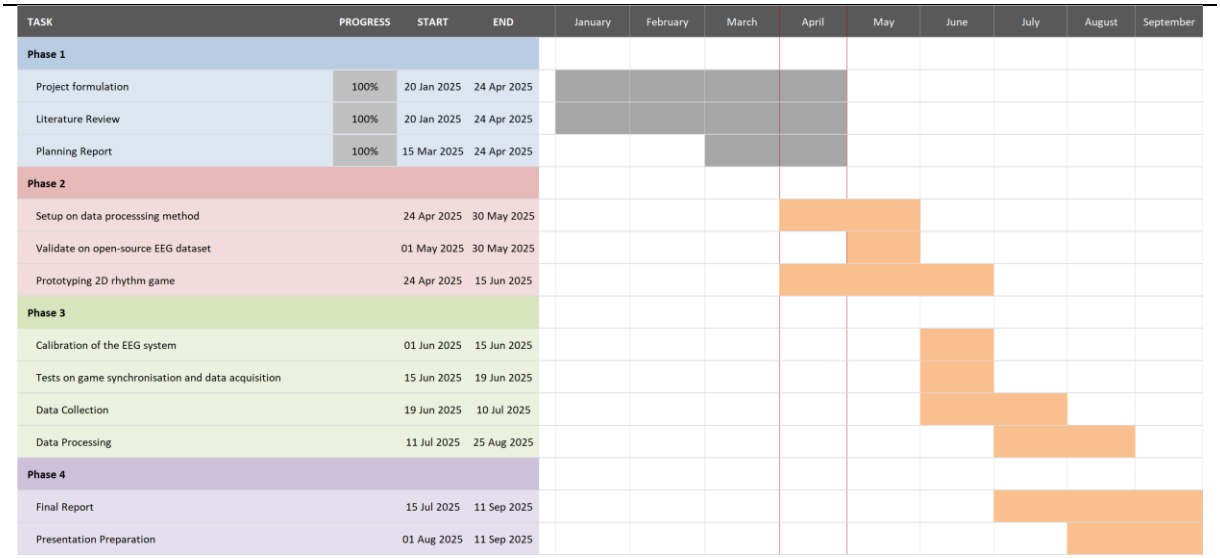

Figure 1: Gantt Chart for the Implementation Plan.

Phase 1: The project is initiated by defining the research focus and outlining key hypotheses. A brief literature review is conducted to establish the framework, and a planning report is prepared.

Phase 2: In this phase, initial data processing methods are validated using an open-source EEG dataset. Simultaneously, a prototype 2D rhythm game is developed in Unity/Godot to deliver synchronized visual and auditory cues that trigger sensorimotor responses. This lays the groundwork for both the processing pipeline and the experimental stimulus.

Phase 3: The project enters its core experimental stage, starting with the calibration of a 16-channel EEG system. Pilot tests are conducted to fine-tune game synchronization and data acquisition. Full-scale data collection follows, and the recorded EEG data is processed and analysed to extract key sensorimotor markers.

Phase 4: In the final phase, findings are consolidated, and a comprehensive report is prepared along with academic and presentation materials.

5. Risk register

Risk	Likelihood	Impact	Mitigation Strategy
Participant Discomfort and Fatigue	Medium	Medium	• Use non-invasive EEG procedures • Schedule regular breaks and monitor participant state during sessions • Provide clear instructions and comfort measures during electrode application
Skin Irritation from Electrodes	Low	Low	• Use high-quality, hypoallergenic electrodes and conductive gel • Pre-screen participants for sensitive skin • Provide immediate treatment or removal in case of irritation
Data Privacy Breach	Low	High	• Anonymize data immediately upon collection • Follow GDPR or equivalent data protection policies
Poor Signal Quality / Motion Artifacts	Medium	High	• Train participants in minimizing movement • Use advanced artifact removal filters and techniques during processing • Pilot testing to fine-tune protocols and adjust electrode placement if needed
Technical Equipment Failure	Low	Medium	• Schedule regular equipment maintenance • Keep backup devices ready • Provide technical support contact information for rapid troubleshooting

Recruitment and Retention Challenges	Medium	Medium	• Offer modest compensation and clear communication regarding participation duration and requirements • Use multiple recruitment channels (e.g., college community, social media)
Financial or Budgetary Constraints	Low	Medium	• Monitor and control expenditures carefully

Table 2: Risk Register. Identify risks, their likelihood, impact, and corresponding mitigation strategy.

6. Evaluation

6.1. Functional Evaluation of the Project Deliverables/Hypotheses

The following table outlines the experimental design used to evaluate each of the project's key hypotheses. For each hypothesis, relevant independent and dependent variables are defined, along with control conditions that serve as baselines for comparison.

Hypothesis	Independent Variable(s)	Dependent Variable(s)/Metrics	Control Condition
H1: Multisensory input enhances and speeds up cortical activation	Type of stimulus (Unisensory vs. Multisensory)	ERD onset latency, SSVEP/ASSR amplitude	Unisensory input (visual-only or auditory-only)
H2: Sequential actions induce more prolonged and distributed ERD	Task type (Single vs. Sequential motor task)	ERD duration and spatial distribution	Single-action motor task
H3: Increased attention improves consistency and timing of brain response	Attention level (High vs. Low)	ERD onset variability (trial-to-trial SD)	Low-attention task (e.g., passive viewing or no-distractor baseline)
H4: Temporal features improve MI-BCI performance	Feature set (With vs. Without temporal features)	Classification accuracy	MI-BCI using non-temporal features (e.g., bandpower only)

Table 3: Summary of hypotheses and the experimental setup (independent variable(s), dependent variable(s)/Metrics, and control condition)

6.2. Technical Performance Testing

Signal Quality Assessment: Measure the electrode impedance, signal-to-noise ratio (SNR), and the prevalence of motion artifacts to ensure data quality.

Temporal Accuracy Testing: Latencies will be measured while running controlled stimulus experiments to verify that the recorded latencies and waveforms match those reported in the literature.

7. Preliminary Results

a) **Baseline Data Integration:** Before collecting EEG data from the multi-sensorimotor experiment, the data processing pipeline is planned to be validated using an open-source EEG dataset. Although no dataset was found that exactly mirrors the planned paradigm for multi-sensorimotor actions, the closest available resource is the "EEG Motor Movement/Imagery Dataset." In this dataset, participants perform or imagine motor tasks in response to visual cues. This resource will serve both to validate the processing methods and as a benchmark for comparing the performance of the MI-BCI classification against established standards. The next step involves developing and validating the data processing pipeline using this dataset.

b) **Experimental Setup Plan:** In parallel, a conceptual design has come up for the experiment using a self-designed 2D rhythm game. In this setup, participants will interact with the game where rhythmic visual cues and/or auditory stimuli signal when to execute specific motor actions.

8. References

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